



## Web Technologies – SEF23

### Homework 09: Real-Time Notifications System Using SignalR

11<sup>th</sup> December, 2025 – Professor Shuja Ur Rehman Baig

#### Important Instructions:

- Any instance of cheating or plagiarism will result in a *F grade*.
- Deadline: 16<sup>th</sup> December, 2025 11:59 PM
- **CLO1:** Know web application architecture and languages & technologies
- **CLO2:** Describe various approaches to web application development
- **CLO3:** Develop web applications

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#### Task Description

Create a real-time notification system in ASP.NET Core MVC using SignalR.

An admin should be able to send a notification message, and all connected users should instantly receive it as a popup/alert.

No notification history is required.

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#### Requirements

##### 1. Models

Model: **Notification**

- string Message
- DateTime Timestamp

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##### 2. Controllers

**Controller Name:** NotificationController

Required action methods:

- Index() → Loads user page and connects to SignalR.
- Admin() → Loads the admin page with a textbox + send button.
- Send(string message) → Sends the notification to the SignalR hub.

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##### 3. SignalR Hub



**Hub Name:** NotificationHub

Must contain:

**Method:** BroadcastNotification(string message) → Sends the message to all connected clients.

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#### 4. Features

- Admin enters a message and clicks Send.
  - All connected clients instantly receive a popup/alert.
  - No saving, no history, no database.
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#### 5. Flow

- User opens Index and connects to the hub.
- Admin opens Admin and sends a message.
- Send() calls the hub.
- Hub broadcasts to all clients.
- Clients show the message as an alert